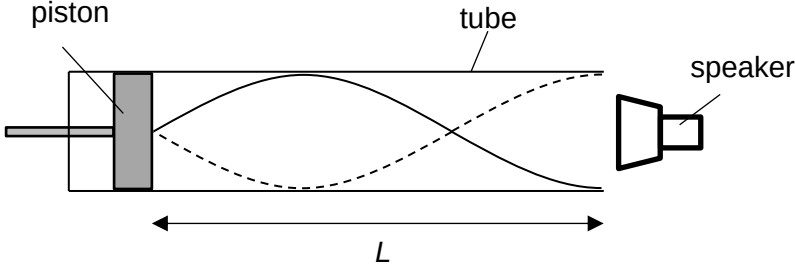


	external device must absorb the lost in G.P.E.	absorb
5(a)(i)	A longitudinal wave is one where the oscillations of the wave particles is parallel to the direction of the wave propagation / energy transfer.	[1]
5(a)(ii)	<u>Sound waves are longitudinal</u> . Since there is <u>no component of a sound wave's oscillation that is perpendicular to its direction of motion</u> , sound waves <u>cannot be polarised</u> .	[1] [1]
5(b)(i)	Sound waves from the speaker travel <u>leftward</u> through the tube and gets <u>reflected</u> by the piston <u>rightwards</u> . The incident sound waves from the speaker <u>superpose</u> with the reflected sound waves by the piston to form stationary waves.	[1] [1]
5(b)(ii)1.		[1]
5(b)(ii)2.	wavelength, $\lambda = \frac{v}{f} = \frac{330}{440} = 0.75 \text{ m}$ $L = \frac{3\lambda}{4} = \frac{3}{4} \times 0.75 = 0.563 \text{ m} = 56.3 \text{ cm}$	[1] for wavelength [1] for correct answer
5(b)(iii)	Stationary waves can only form at specific lengths of the air column, such that <u>a node and an antinode are formed at the closed and opened end of the tube respectively</u> .	[1] [1]

	This can only occurs when <u>L is odd number multiples of quarter wavelength</u> or $L = (\text{odd number}) \frac{\lambda}{4}$.	
6(a)		[1] show at least